

DATE : 20.12.2024
DT/NT : DT
LESSON : LINUX
SUBJECT: USING DIRECTORIES
LISTING FILES
SESSION : 4
BATCH : B 303

AWS-DEVOPS



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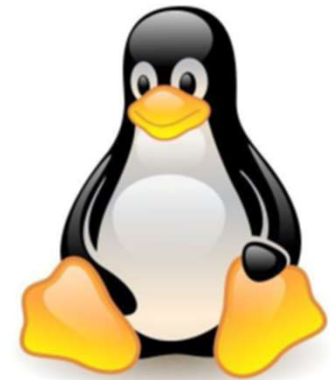
- ▶ Using Directories Listing Files
- ▶ Package Management
- ▶ Popular Linux System Package Managers





Using Directories

Listing Files



Files and Directories



- The file system hierarchy standard (FHS) defines the structure of the file systems on Linux.
- In the FHS, all files and directories appear under the root directory /, even if they are stored on different physical or virtual devices.
- Most of these directories exist in all UNIX, however, they are not considered authoritative for platforms other than Linux.

Files and Directories



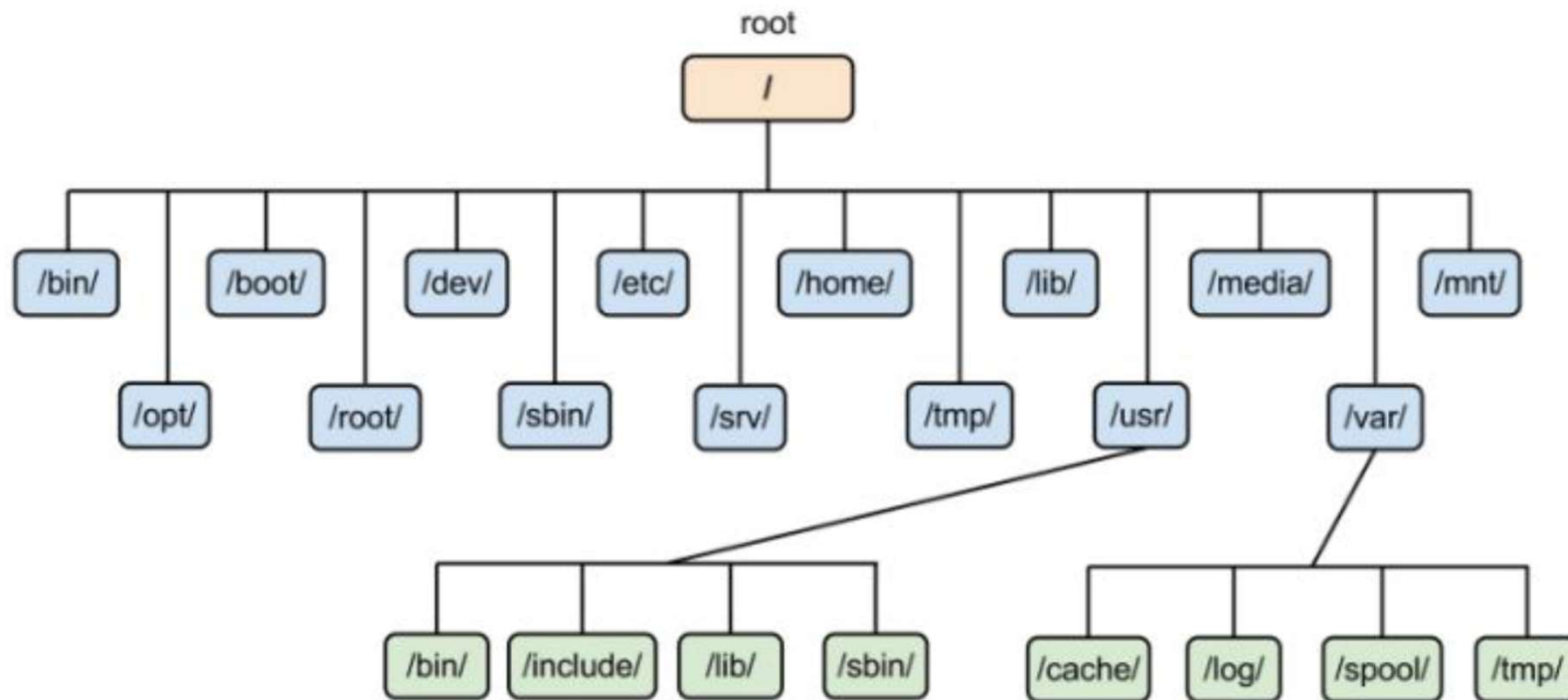
- On a Linux system, everything is a file.
- A Linux system makes no difference between a file and a directory, since a directory is just a file containing names of other files.
- The tree of the file system starts at the trunk or slash, indicated by a forward slash (/). This directory, containing all underlying directories

Files and Directories

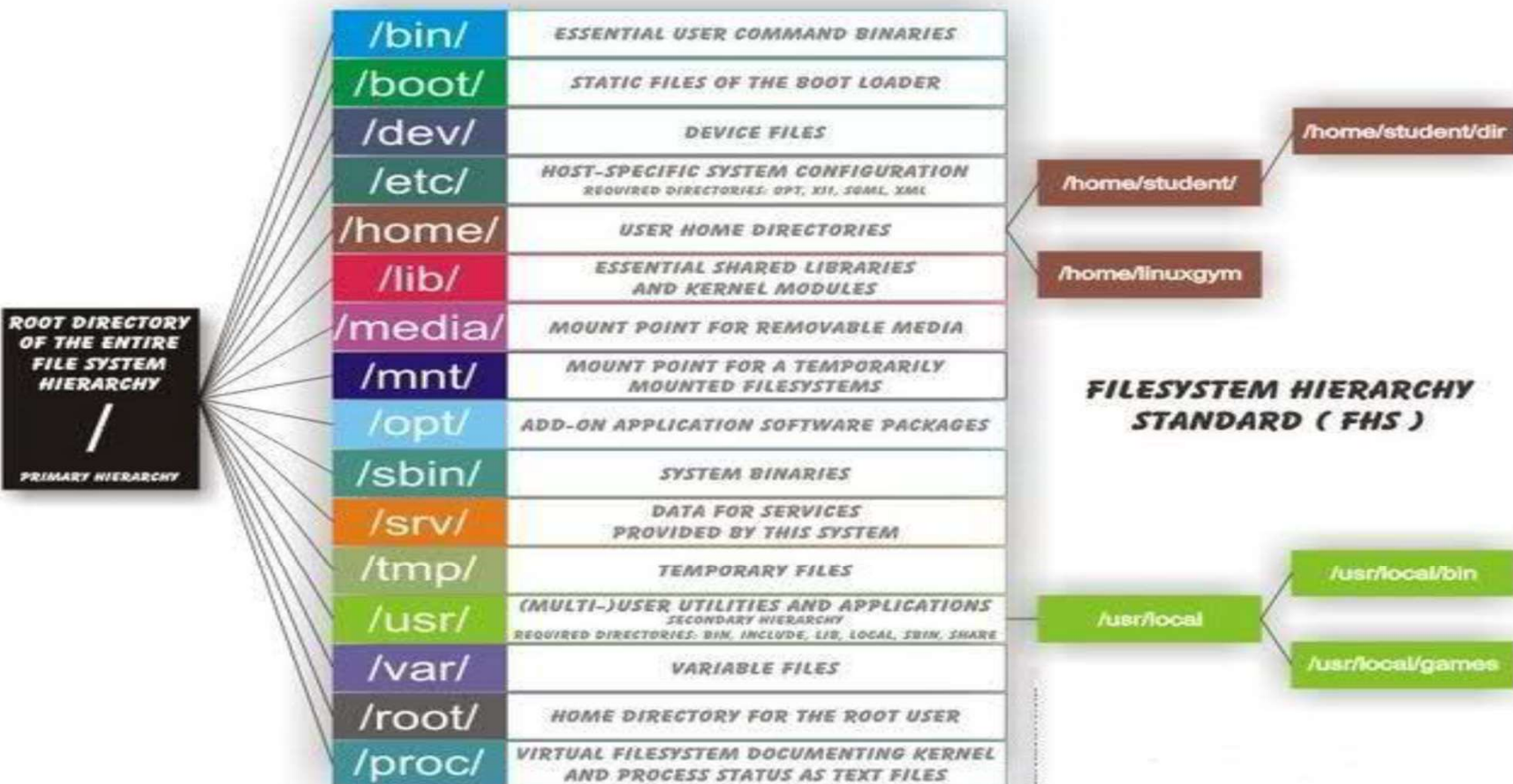


<code>/root</code>	• Home directory of the root user
<code>/bin</code>	• Essential command binaries
<code>/boot</code>	• Boot loader files
<code>/dev</code>	• Essential device files
<code>/etc</code>	• Host-specific configuration files
<code>/home</code>	• Users' home directories
<code>/lib</code>	• Libraries essential for the binaries
<code>/mnt</code>	• Temporarily mounted filesystems.
<code>/opt</code>	• Optional application packages
<code>/proc</code>	• Contains information about system
<code>/sbin</code>	• Essential system binaries
<code>/tmp</code>	• Temporary files
<code>/var</code>	• Variable data files

Files and Directories



Files and Directories



Files and Directories



Symbol	Meaning
-	Regular file
d	Directory
l	Link
c	Character Device File
s	Socket File
p	Named Pipe
b	Block Device

-rw-----	Regular File
d rw-r-xr-x.	Directory File
l rw-rwxrwx.	Link File
c rw-rw----.	Character Device File
b rw-rw----.	Block Special File
s rw-rw-rw-	Socket File
p rw-----.	Named Pipe File

Files and Directories

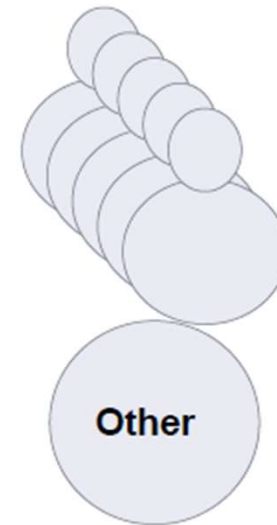
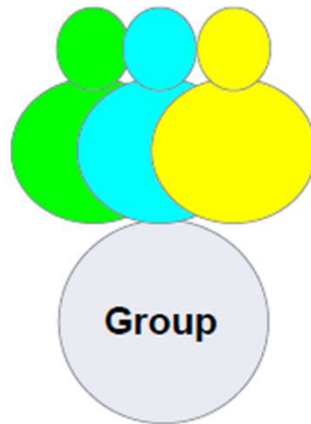
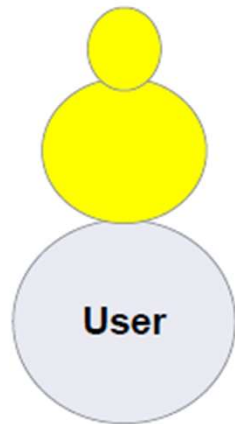


- On most Linux versions ls is aliased to color-ls by default. This feature allows to see the file type without using any options to ls.

Color	Meaning
Blue	directories
Red	compressed archives
White	text files
Pink	images
Cyan	links
Yellow	Devices
Green	Executables
flashing red	broken links

```
linuxuser@techproeducation: ~$ cd lib
linuxuser@techproeducation: /lib$ ls
apparmor      initramfs-tools  recovery-mode
apt            kernel           rsyslog
binfmt.d      locale           sasl2
byobu          lsb              snapd
cnf-update-db man-db           software-properties
command-not-found mime              ssl
compat-ld     modprobe.d       sysctl.d
console-setup modules-load.d    systemd
dbus-1.0      netplan          sysusers.d
dpkg          networkd-dispatcher tc
environment.d openssh           terminfo
file          os-release       tmpfiles.d
girepository-1.0 pam.d            ubuntu-advantage
git-core      pm-utils         ubuntu-release-upgrader
gnupg         policykit-1      udev
gnupg2        polkit-1         ufw
gold-ld       python2.7        usrmerge
groff         python3           valgrind
hdparm        python3.10       wsl
init          python3.11       x86_64-linux-gnu
linuxuser@techproeducation: /lib$
```

File Permission



Ownership



File Permission



Read



Write



Execute



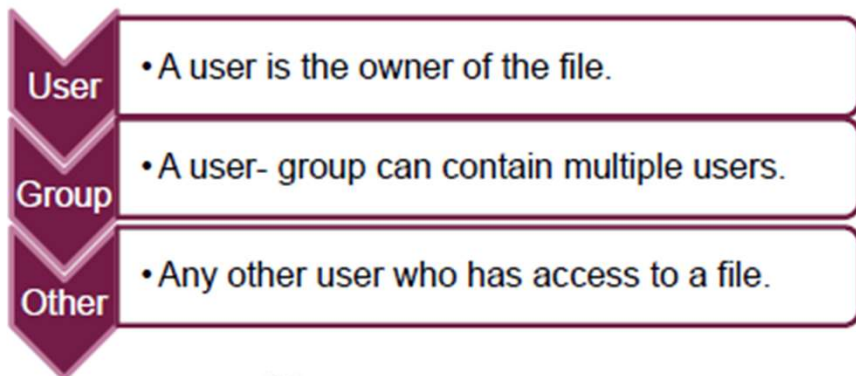
Ownership



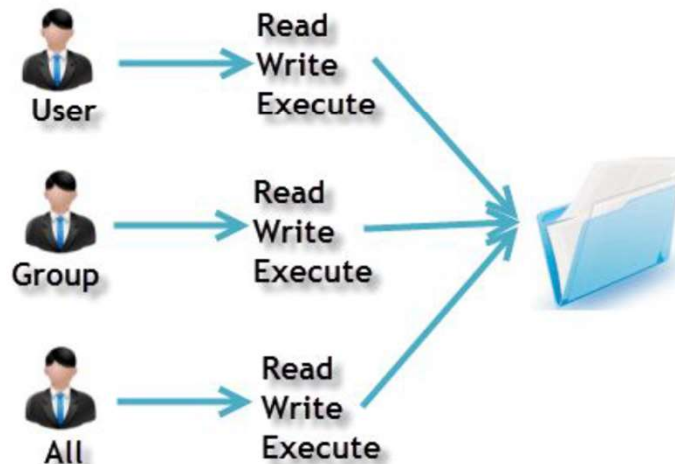
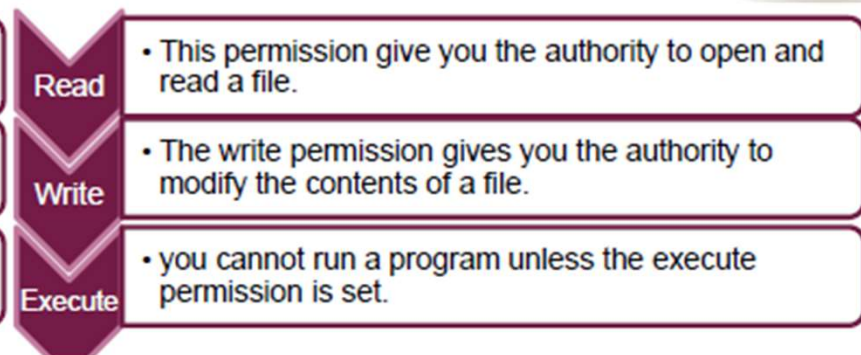
File Permission



Ownership



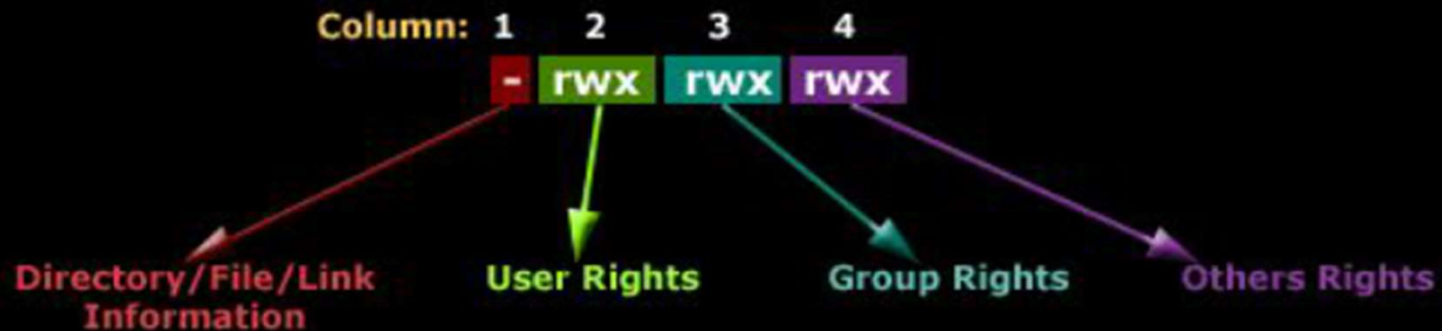
Permission



File Permission

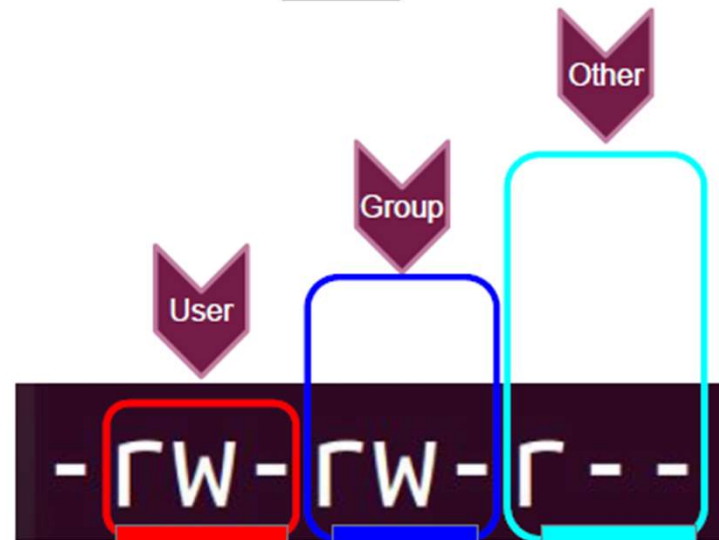


Understanding The Linux File Permissions



While the first column defines a directory, file or link, the next 3 columns (2, 3, 4) define the permissions for the User, Group and Others (everyone else) groups.

File Permission



```
-rw-rw-r-- 1 zk zk 0 Dec 7 15:39 html.txt
```

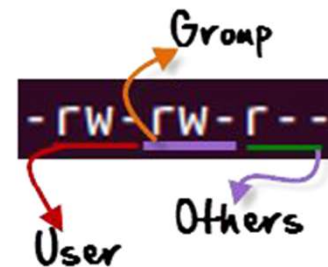


Ownership

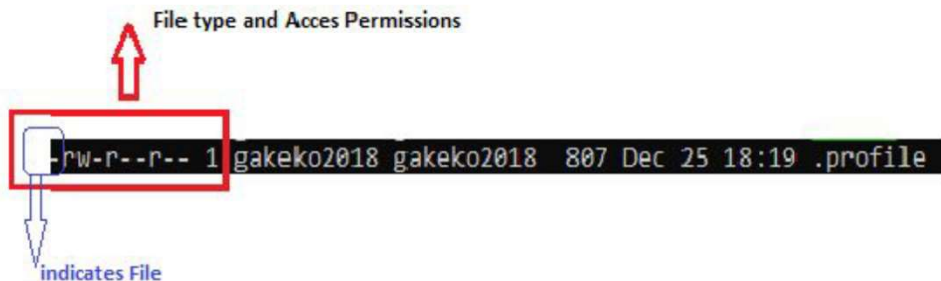


File Permission

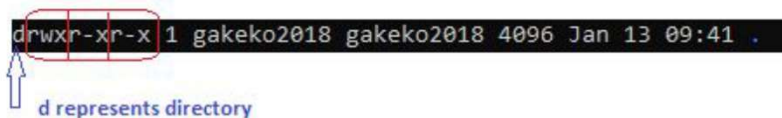
```
linuxuser@techproeducation: X + v
linuxuser@techproeducation:~$ ls -l
total 21260
-rw-rw-r-- 1 linuxuser linuxuser 21765972 Nov 30 11:03 latest-hugo.zip
drwxrwxr-x 2 linuxuser linuxuser 4096 Nov 30 09:56 linux_lesson
linuxuser@techproeducation:~$ |
```



r: Read
w: Write
x: Execute



r = read permission
w = write permission
x = execute permission
- = no permission



File Permission



Changing Permission with chmod Command

- We can use the chmod command which stands for change mode.
- we can set permissions (read, write, execute) on a file/directory for the owner, group and the world.

`chmod permissions filename`

`chmod u=rwx,g=rx,o=r myfile`

Symbol	Permission Type
---	No Permission
--x	Execute
-w-	Write
-wx	Execute+Write
r--	Read
r-x	Read+Execute
rw-	Read+Write
rwx	Read+Write+Execute

File Permission



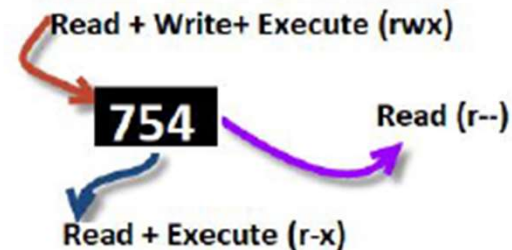
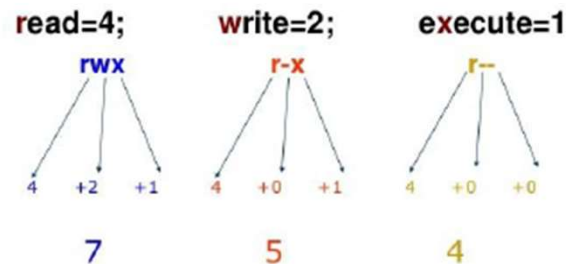
```
linuxuser@techproeducation: ~  
linuxuser@techproeducation:~/permission$ ls -l  
total 0  
-rw-rw-r-- 1 linuxuser linuxuser 0 Dec 1 10:40 myfile  
linuxuser@techproeducation:~/permission$ chmod 754 myfile  
linuxuser@techproeducation:~/permission$ ls -l  
total 0  
-rwxr-xr-- 1 linuxuser linuxuser 0 Dec 1 10:40 myfile  
linuxuser@techproeducation:~/permission$
```

754 code says;

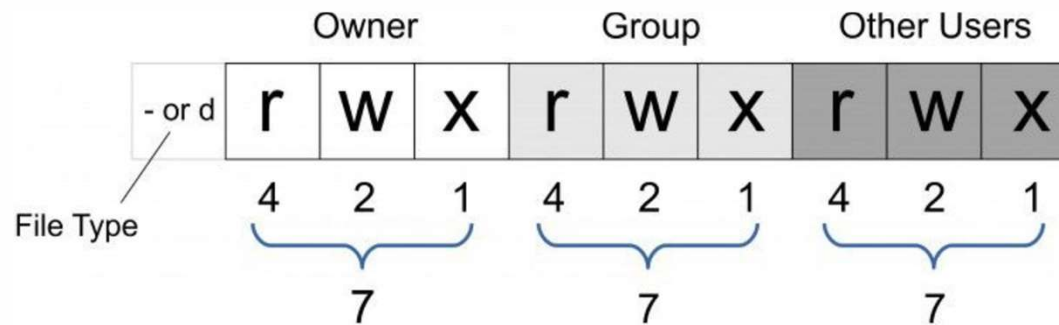
- Owner can read, write and execute
- User's group can read and execute
- Other can only read

```
chmod u=rwx,g=rx,o=r myfile  
chmod 754 myfile
```

Permissions



File Permission



d	r	w	x	r	-	x	r	-	-
	read	write	exec	read	write	exec	read	write	exec
File type	Owner permissions			Group permissions			User permissions		
(directory)	4	2	1	4	2	1	4	2	1
	7			5			4		

File Permission



File Attributes	Meaning
-rwx-----	A regular file that is readable, writable, and executable by the file's owner. No one else has any access.
-rw-----	A regular file that is readable and writable by the file's owner. No one else has any access.
-rw-r--r--	A regular file that is readable and writable by the file's owner. Members of the file's owner group may read the file. The file is world-readable.
-rwxr-xr-x	A regular file that is readable, writable, and executable by the file's owner. The file may be read and executed by everybody else.
-rw-rw----	A regular file that is readable and writable by the file's owner and members of the file's group owner only.
lrwxrwxrwx	A symbolic link. All symbolic links have "dummy" permissions. The real permissions are kept with the actual file pointed to by the symbolic link.
drwxrwx---	A directory. The owner and the members of the owner group may enter the directory and create, rename and remove files within the directory.
drwxr-x---	A directory. The owner may enter the directory and create, rename, and delete files within the directory. Members of the owner group may enter the directory but cannot create, delete, or rename files.

File Permission



Octal	Binary	File Mode
0	000	---
1	001	--x
2	010	-w-
3	011	-wx
4	100	r--
5	101	r-x
6	110	rw-
7	111	rwX

By using three octal digits, we can set the file mode for the owner, group owner, and world.

File Permission



Notation	Meaning
u+x	Add execute permission for the owner.
u-x	Remove execute permission from the owner.
+x	Add execute permission for the owner, group, and world. This is equivalent to a+x.
o-rw	Remove the read and write permissions from anyone besides the owner and group owner.
go=rw	Set the group owner and anyone besides the owner to have read and write permission. If either the group owner or the world previously had execute permission, it is removed.
u+x, go=rx	Add execute permission for the owner and set the permissions for the group and others to read and execute. Multiple specifications may be separated by commas.

File Permission



Set permissions of myfile.txt to;

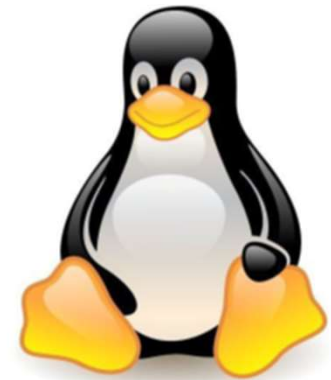
owner : full access

group : read and execute

others : no access



Package Management



Package Management



A package manager is a collection of software tools that automates the process of installing, upgrading, configuring, and removing computer programs for a computer's operating system in a consistent manner.



Package Manager



Office Boy

Package Management



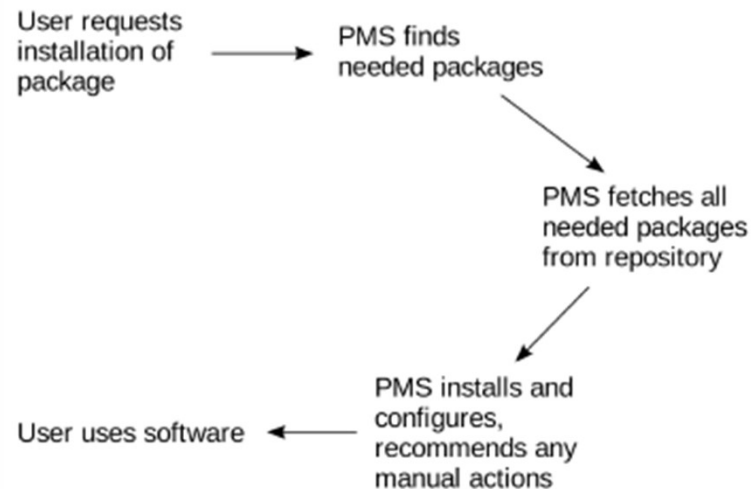
A package manager deals with packages, distributions of software and data in archive files. Packages contain metadata, such as the software's name, description of its purpose, version number, vendor, checksum, and a list of dependencies necessary for the software to run properly. Upon installation, metadata is stored in a local package database.



Package Management



Package managers are designed to eliminate the need for manual installs and updates. This can be particularly useful for large enterprises whose operating systems are typically consisting of hundreds or even tens of thousands of distinct software packages.



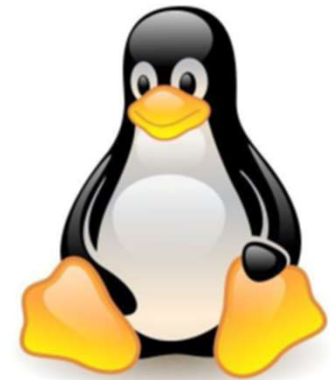
Package Management



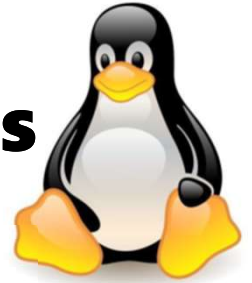
Operating System	Format	Tool(s)
Debian	.deb	apt , apt-cache , apt-get , dpkg
Ubuntu	.deb	apt , apt-cache , apt-get , dpkg
CentOS	.rpm	yum
Fedora	.rpm	dnf
FreeBSD	Ports, .txz	make , pkg



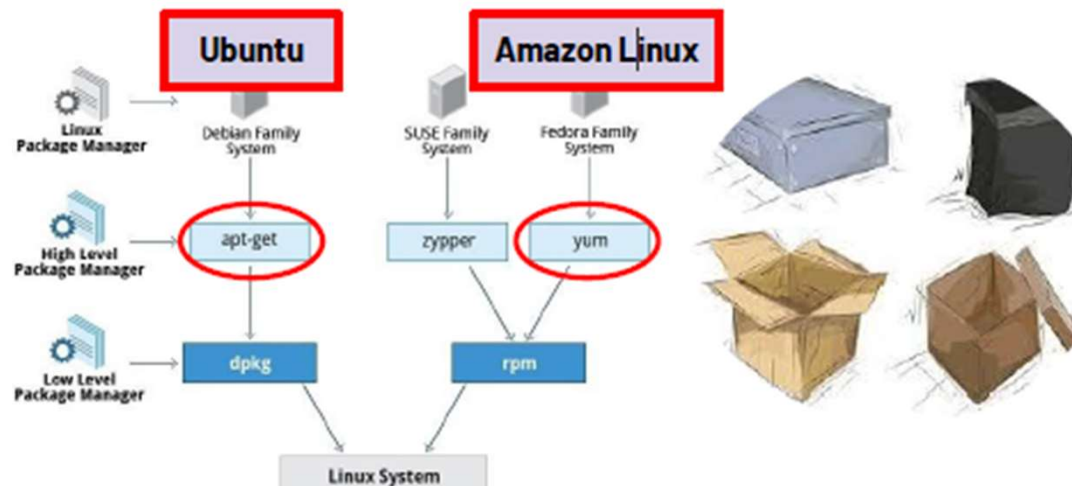
Popular Linux System Package Managers



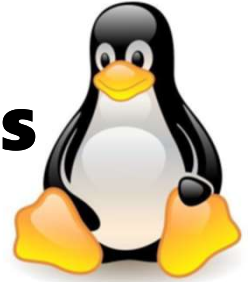
Popular Linux System Package Managers



Linux systems use package managers to add or remove the software packages. These package managers are also a package so you can install any of them.



Popular Linux System Package Managers

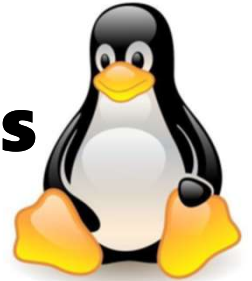


Debian Package Managers

dpkg is the main package management program for the Debian Linux distros. It is used to handle Debian package files with the extension of .deb

```
$ dpkg -i [package-name] # Installing a package
$ dpkg -r [package-name] # Removing a package
$ dpkg -l                 # Lists installed packages
```

Popular Linux System Package Managers



Debian Package Managers

APT (Advanced Package Tool)

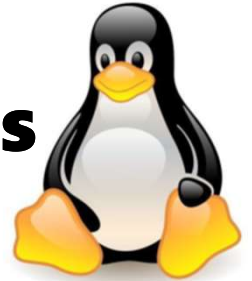
- The **A**dvanced **P**ackaging **T**ool is what Ubuntu Software Center is built on



- 'apt-get install PACKAGE' will install and organize software
- 'apt-cache list PACKAGE' will search for PACKAGE in the local database
- 'apt-get update' update the local package database

```
$ apt update                # Update the installed packages
$ apt install [package-name] # Install a package and all its dependencies
$ apt remove [package-name] # Remove a package
$ apt purge [package-name]  # Remove a package and its configuration files
```


Popular Linux System Package Managers



Debian Package Managers

➔ Aptitude Package Manager

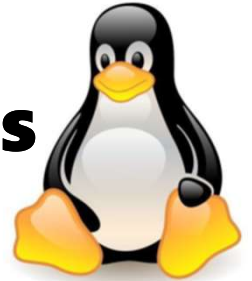
aptitude tool provides the functionality of **apt-get**, as well as many additional features:

- aptitude provides easy access to all versions of a package
- aptitude tracks of obsolete software
- aptitude has a powerful system for searching particular packages

```
$ aptitude install [package-name] # Install a package  
$ apt-get install [package-name] # Install a package
```



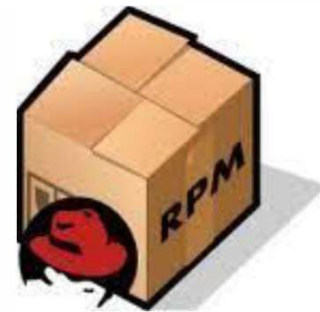
Popular Linux System Package Managers



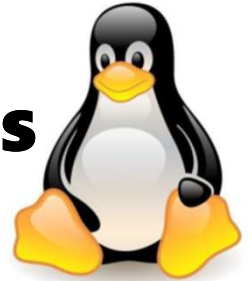
Red Hat Package Managers

rpm is the package manager for Red Hat Linux operating systems. The installation package files have .rpm extension. These files are used for installing programs. rpm command has been used for RPM packages by default but new tools are developed for better performance.

```
$ rpm -i [package-name] # Install a package  
$ rpm -e [package-name] # Uninstall a package
```



Popular Linux System Package Managers



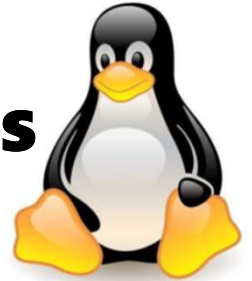
Red Hat Package Managers

YUM is an open-source package manager that was developed by Duke University. It is used both in the command line and GUI. It works mostly the same as APT in Debian Linux systems. Here are some examples of YUM.

```
$ yum install [package-name]    # Install a package
$ yum remove [package-name]    # Remove a package
$ yum update [package-name]    # Update a package
```



Popular Linux System Package Managers



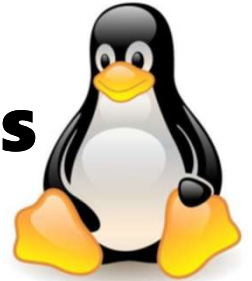
Red Hat Package Managers

➡ DNF – Dandified Yum

It is the new generation of YUM package manager. It is the default package manager of Fedora 22 and newer distros. The usage of DNF is mostly the same as YUM..

```
$ yum install dnf    # Install DNF via yum.  
$ dnf --version      # Checking DNF version  
$ dnf install         # Installing a package
```

Popular Linux System Package Managers



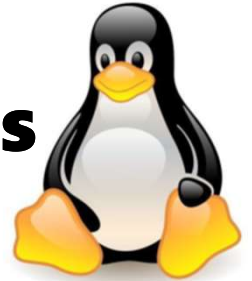
Red Hat Package Managers



Other RPM tools:

- zypper (openSUSE)
- up2date (Red Hat Enterprise Linux, CentOS 3 and 4, and Oracle Linux)
- urpmi (Mandriva Linux, ROSA Linux, and Mageia)
- apt-rpm (Ark Linux,[11] PCLinuxOS and ALT Linux)
- smart (Unity Linux and Fedora)
- rpmquery (Red Hat Enterprise Linux)

Popular Linux System Package Managers



Other Package Managers

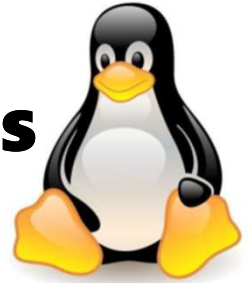


Below are a few more notable/interesting package managers.

- Portage: Package manager for Gentoo.
- Pacman: Arch Linux Package manager.
- Nix: A 'Fully Functional/Transactional' package manager.
- Brew: An Open Source package manager for OSX.**
- Chocolatey: A package manager for Windows.**



Popular Linux System Package Managers



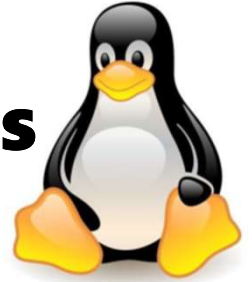
Other Package Managers

Programming languages have their own default package managers. They help to find and install the packages via searching libraries that exist on the internet for that language.

Examples: **Python: pip** / Ruby: gem, rubygems / Haskell: cabal / NodeJS: npm



Popular Linux System Package Managers



- `yum install [package-name]` → Bir paket yükle
- `yum -y install [package-name]` → Kurulum sırasında onayları atla
- `yum remove [package-name]` → Bir paketi kaldırın.
- `yum autoremove [package-name]` → Bir paketi ve kullanılmayan bağımlılıkları kaldırın
- `yum update [package-name]` → Bir paketi güncelle
- `yum update` → Kurulu tüm paketleri güncelle
- `yum info [package-name]` → Bir paket hakkında bilgi alın
- `yum list` → Kurulu ve mevcut tüm paketleri listele
- `yum list [package-name]` → Mevcut eşleşen paket(ler)i listele
- `yum list installed` → Kurulu paketleri listele
- `yum --showduplicates list [package-name]` → Mevcut tüm sürümleri listeler
- `yum install [package-name]-[version]` → Belirli bir sürümü yükle

Exercise-1

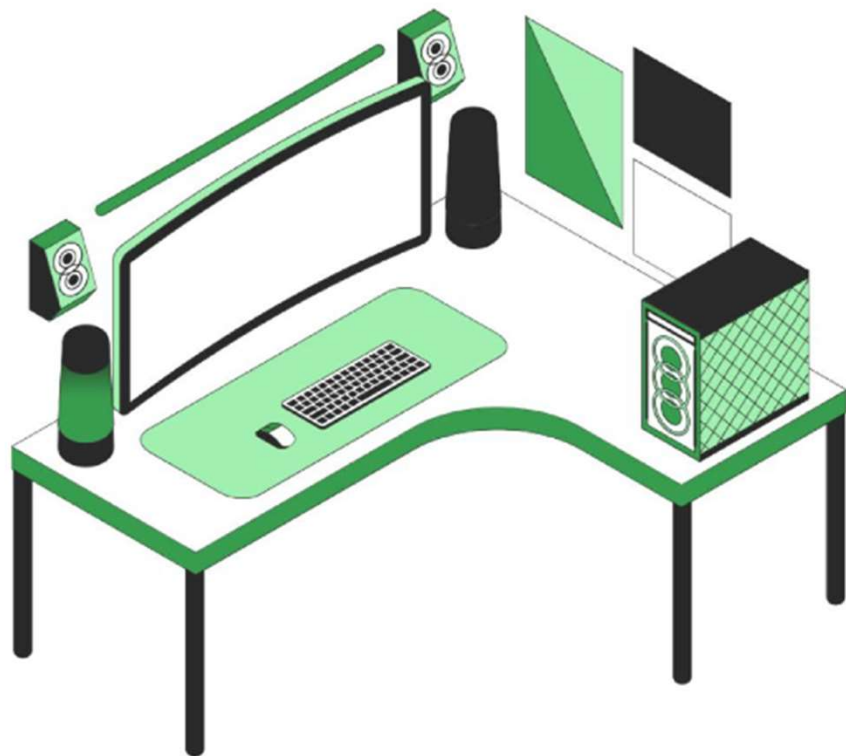


1. Kurulu tüm paketleri **güncelle**
2. **http** ile başlayan tüm kurulu paketleri **listele**
3. **http** ile başlayan mevcut tüm paketleri bulun
4. Varsa **httpd'yi** yükleyin. (Kurulum sırasında onayları atlayın)
5. Kurulu **httpd** paketini **listele**
6. **httpd'yi kaldır**
7. Kurulu **httpd** paketini **listele**

Exercise-2



1. **Git'i** kullanılmayan tüm bağımlılıklarla birlikte **kaldırın**
2. Kurulu **git'i** kontrol et
3. Kullanılabilir önceki **git** sürümünü bulun
4. Mevcut önceki **git** sürümünü yükle
5. Yüklü **git** sürümünü kontrol edin
6. **Git'i** en son sürüme güncelleyin
7. Yüklü **git** sürümünü kontrol edin



Do you have any questions?

Send it to us! We hope you learned something new.